Akash Singh Sagar

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Summary

Computer Science Engineering undergraduate specializing in Data Science, with hands-on experience building machine learning, NLP, generative AI, and full-stack applications. Experienced in developing AI agents, ML pipelines, REST APIs, recommendation systems, and predictive applications using Python, FastAPI, React.js, LangChain, and Mistral AI, backed by a strong foundation in data structures, algorithms, and systems fundamentals.

Education

K.R. Mangalam University — Gurugram, India
Bachelor of Technology in Computer Science Engineering – Data Science — *Expected Graduation: 2027*

Technical Skills

Languages: Python, SQL, Java, JavaScript, R
Machine Learning & Data Science: Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, Feature Engineering, EDA, Model Evaluation
NLP & Generative AI: TF-IDF, NLTK, LangChain, Mistral AI, Large Language Models, RAG, Multi-Agent Systems
Backend & APIs: FastAPI, REST APIs, CRUD Operations
Frontend: React.js, HTML, CSS
Visualization: Power BI, Tableau, Matplotlib, Seaborn, Microsoft Excel
Computer Science Fundamentals: Data Structures & Algorithms, Operating Systems, Computer Networks, OOP
Tools & Deployment: Git, GitHub, Docker, Streamlit, Jupyter Notebook, Google Colab, Render, Vercel, Railway

Experience

Software Developer Intern
AattamNirvana
- Built reusable React.js components for responsive web applications, improving UI consistency and reducing duplicated frontend code.
- Implemented Odoo CRM API integration to support customer data management and CRUD-based application workflows.
- Connected Calendly APIs with application workflows to enable appointment scheduling.
- Developed and configured REST APIs to enable communication between frontend applications and external services.
- Collaborated within an Agile development workflow to implement features, troubleshoot issues, and improve application functionality.

National-Level Hackathon Participant – Code Xccelerator
Gateway Education
- Won the Best Social Impact Award at a 72-hour national-level hackathon, competing against 70+ teams from across India.
- Developed a social-impact-focused technology solution under a tight development timeline.
- Contributed to rapid prototyping, technical problem solving, and software development in a high-pressure team environment.

Projects

AI Interviewer | *FastAPI, React.js, Mistral AI, LangChain, Python*
- Built a full-stack AI-powered technical interview platform with five REST API endpoints and five specialized AI agents for personalized technical interviews generated from candidate resumes and previous responses.
- Implemented Resume Analysis, Interviewer, Live Interviewer, Answer Evaluator, and Final Report agents using Mistral AI (mistral-large-latest) and LangChain.
- Designed a five-stage interview pipeline with two questions per stage, generating ten personalized questions based on candidate resumes and prior responses.
- Developed FastAPI services for resume upload, PDF text extraction, candidate profiling, interview sessions, answer evaluation, and final reporting.
- Implemented structured evaluation across relevance, technical knowledge, clarity, completeness, and confidence, with automated performance reporting.

ResearchX – AI Multi-Agent Research System | *Python, LangChain, Mistral AI, Tavily, Streamlit*
- Built an agentic AI research platform using four specialized AI agents across a five-stage research pipeline.
- Implemented Search, Reader, Writer, and Critic agents to automate web search, information extraction, report generation, and quality evaluation.
- Integrated Mistral AI, LangChain, and the Tavily search API for API-based information retrieval, analysis, synthesis, and review.
- Developed a structured workflow that transforms user queries into organized research reports, with a dedicated Critic Agent for identifying information gaps.
- Built an interactive Streamlit interface for submitting research queries and monitoring the research workflow.

GridCast – Smart Load Prediction | *Python, FastAPI, React.js, XGBoost, TensorFlow, Scikit-learn*
- Built an end-to-end electricity demand forecasting system using real-world Northern Regional Load Despatch Centre (NRLDC) electricity data for short-term power-load prediction.

- Automated data ingestion and preprocessing for 15-minute interval electricity demand forecasting with a 24-hour prediction horizon.
- Implemented XGBoost-based forecasting and developed FastAPI REST APIs for serving demand predictions.
- Built a React.js dashboard for visualizing forecasts, demand patterns, and residual heatmaps.
- Achieved 1.8% MAPE for 24-hour-ahead electricity demand forecasting.

Movie Recommendation System | *Python, Scikit-learn, FastAPI, React.js, Tailwind CSS*

- Built a full-stack content-based movie recommendation system using TF-IDF vectorization and cosine similarity.
- Processed 45,467 movie records across 24 attributes and engineered textual metadata for TF-IDF-based content similarity.
- Developed FastAPI REST APIs for movie search, movie details, content-based recommendations, and genre-based fallback recommendations.
- Integrated external movie APIs to provide posters, ratings, release dates, genres, backdrops, and additional metadata.
- Developed a responsive React.js and Tailwind CSS frontend with autocomplete search, movie details, and recommendation sections.

Student Grade Prediction System | *Python, FastAPI, React.js, Random Forest, Scikit-learn*

- Built a full-stack ML application to predict student final grades using a Random Forest Regressor configured with 100 trees.
- Trained the model on 395 student records with 32 features using an 80/20 train-test split.
- Achieved an R^2 of 0.8121 with a mean absolute error (MAE) of 1.1585 grade points.
- Developed a FastAPI REST API and React.js dashboard for model predictions and performance visualization.

Certifications

- Microsoft Certified: Azure Data Fundamentals (DP-900) – Microsoft, August 2025
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle, August 2025
- Deloitte Data Analytics Job Simulation Certificate – Deloitte, March 2026
- Machine Learning Certification – K.R. Mangalam University, in association with IBM
- Data Analysis Certification – [Add issuing platform/organization]
- Data Visualization Certification – [Add issuing platform/organization]
- R Programming Certification – [Add issuing platform/organization]